

Dhawal Sirikonda

Ph.D. Candidate • Rendering & Imaging Science Lab (RISc), Dartmouth College

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🔬 research summary

I build next-generation ultra-fast LiDAR sensors and computational imaging systems for high-speed perception and communication. My work designs sensing architectures that reach MHz-rate depth measurements in time-of-flight sensors, and uses ultrafast acousto-optic lensing for high-speed structured-light scanning ($\sim 1000\times$ faster than prior art) and underwater optical backscatter communication ($\sim 600\times$ faster). I combine learning-based and physics-driven methods across event cameras, RGB-D, and emerging hybrid sensing modalities. I am advised by [Prof. Adithya Pediredla](#). Before my Ph.D. I completed an M.S. at IIIT-Hyderabad with [Prof. P. J. Narayanan](#) on ML-driven graphics and 3D vision.

🎓 education

Sep 2023 – Present	Ph.D., Computer Science — Dartmouth College Rendering & Imaging Science Lab (RISc) • Advisor: Prof. Adithya Pediredla	Hanover, NH, USA
Jan 2020 – May 2023	M.S., Computer Science — IIIT-Hyderabad Center for Visual Information Technology (CVIT) • Advisor: Prof. P. J. Narayanan Thesis: Real-time Rendering of Arbitrary Surface Geometries using Precomputed Radiance Transfer	Hyderabad, India
Aug 2019 – Dec 2019	M.Tech., Computer Science (discontinued) — IIIT-Hyderabad Discontinued after one semester to transfer into the M.S. by Research program	Hyderabad, India
2014 – 2018	B.Tech., Computer Science — JNTUK – University College of Engineering Academically 2nd in the 2014–2018 batch (Roll of Honours)	Vizianagaram, India

📖 publications

- [1] [SDR-ToF: Million Optical Depth Samples per Second Using Software-Defined Radio](#) *SIGGRAPH Asia (ACM ToG)*, 2026.
Ramchander Bhaskara*, **Dhawal Sirikonda***, Omkar Vengurlekar*, Juhyeon Kim, Joseph Lazarro, Suren Jayasuriya, Adithya Pediredla (* $\equiv$ joint first authors)
- [2] [Underwater Optical Backscatter Communications using Acousto-Optic Beam Steering](#) *SIGGRAPH Asia (ACM ToG)*, 2025.
Atul Rohit Agarwal*, **Dhawal Sirikonda***, Atharv Agashe, Ziang Ren, Dinithi Silva-Sassaman, Charles Carver, Alberto Quattrini Li, Xia Zhou, Adithya Pediredla (* $\equiv$ joint first author)
- [3] [Structured Light with a Million Light Planes per Second](#) *ICCP & TPAMI*, 2025.
Dhawal Sirikonda, Praneeth Chakravarthula, Ioannis Gkioulekas, Adithya Pediredla
- [4] [GSN: Generalisable Segmentation in Neural Radiance Field](#) *AAAI*, 2024.
Vinayak Gupta, Rahul Goel, **Dhawal Sirikonda**, P. J. Narayanan
- [5] [FusedRF: Fusing Multiple Radiance Fields](#) *CVPR Workshop (XRNeRF)*, 2023.
Rahul Goel, **Dhawal Sirikonda**, Rajvi Shah, P. J. Narayanan
- [6] [Interactive Segmentation of Radiance Fields](#) *CVPR*, 2023.
Rahul Goel*, **Dhawal Sirikonda***, Saurabh Saini, P. J. Narayanan (* $\equiv$ joint first author)
- [7] [Real-Time Rendering of Arbitrary Surface Geometries using Learnt Transfer](#) *ICVGIP*, 2022.
Dhawal Sirikonda, Aakash KT, P. J. Narayanan
- [8] [StyleTRF: Stylizing Tensorial Radiance Fields](#) *ICVGIP*, 2022.
Rahul Goel*, **Dhawal Sirikonda***, Saurabh Saini, P. J. Narayanan (* $\equiv$ joint first author)

[9] Neural View Synthesis with Appearance Editing from Unstructured Images

ICVGIP, 2021.

Pulkit Gera, Aakash KT, **Dhawal Sirikonda**, P.J. Narayanan

preprints & posters

- › **Learnt Transfer for Surface Geometries** HPG 2022 (Poster)
- › **Transfer Textures for Fast Precomputed Radiance Transfer** EuroGraphics 2022 (Poster)
- › **PRTT: Precomputed Radiance Transfer Textures** arXiv 2022
- › **Appearance Editing with Free-viewpoint Neural Rendering** arXiv 2021

🌟 patents

- › **MHz-Rate Optical Depth Imaging Using Software-Defined Radios**
U.S. Patent App. No. 64/113,861, filed Jul. 17, 2026 • Pediredla et al.
- › **Underwater Optical Backscatter Communication Using Acousto-Optic Beam Steering**
U.S. Patent App. No. 63/913,085, filed Nov. 6, 2025 • Pediredla et al.
- › **Event-Enhanced Histopathology Apparatus**
U.S. Patent App. No. 63/837,253, filed Jul. 2, 2025 • Pediredla et al.

🎤 talks & presentations

- › Using Sound to Steer Light and Light to Measure Sound VASC Seminar, Robotics Institute, Carnegie Mellon University • 2026
- › [live demo] Controlling Light With Sound ICCP 2026, Princeton, NJ, USA • 2026
- › From Rendering to Imaging: Controlling Light with Sound NESG 2026, New England Symposium on Graphics, MIT • 2026
- › Underwater Optical Backscatter Communications using Acousto-Optic Beam Steering SIGGRAPH Asia 2025, Hong Kong SAR • 2025
- › Structured Light with a Million Light Planes per Second NECV 2025, University of Massachusetts Amherst • 2025
- › Structured Light with a Million Light Planes per Second ICCP 2025, Toronto, Canada • 2025
- › Interactive Segmentation of Radiance Fields NECV 2023, Dartmouth College • 2023

🧰 experience

Sep 2023 – Present	PhD Researcher <i>Rendering & Imaging Science Lab, Dartmouth College</i> Research on acousto-optic imaging for ultra-fast scanning, optical communication, and acoustic lensing; ultra-fast low-cost LiDAR from commodity RF hardware.	Hanover, NH, USA
June 2026 – Sept 2026	Machine Learning Engineer <i>Bosch Research</i>	Pittsburgh, PA, USA
Jan 2020 – May 2023	Research Assistant <i>Center for Visual Information Technology, IIIT-Hyderabad</i> Differentiable rendering and multi-view surface reconstruction (Mitsuba 2); supervised undergraduate and dual-degree students.	Hyderabad, India

👤 teaching

Dartmouth College | Teaching Assistant

Computational Photography	Spring 2024 & 2025
Computer Vision	Winter 2024

IIIT-Hyderabad | Teaching Assistant

Advanced Graphics (AR/VR)	Fall 2022
Computer Graphics	Spring 2021

TalentSprint | Mentor — AI/ML program (with IIIT-Hyderabad)

Capstone projects for working professionals	2022 – 2023
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mentoring

- › Ava Carlson Undergraduate, Dartmouth College 2025 – Present
- › Saurish Gali High-school student, Oakton High School 2025
- › Atul R. Agarwal Graduate student, Dartmouth College 2024 – 2025
- › Atharv Agashe Undergraduate, Dartmouth College 2024 – 2025
- › Rahul Goel Undergraduate, IIIT-Hyderabad 2022 – 2023

honours & service

- › **Reviewer:** ICVGIP 2023; AAAI 2026; CVPR 2026; SIGGRAPH 2026; NeurIPS 2026 2023 – Present
- › **Roll of Honours** — academically 2nd in the 2014–2018 batch, JNTUK–UCEV 2018
- › Certified Programmer in Building Systems and Applications, MissionRnD 2016 – 2017

outreach

2024 – 2026 Dartmouth Science Day
Demos to spark grade 2–10 students’ curiosity about imaging science; Hanover, NH.

technical skills

Programming Python, C/C++, MATLAB

Libraries & Frameworks PyTorch, CUDA, OptiX, OpenGL, Mitsuba 2

Focus areas Computational imaging, physically based rendering, advanced graphics (AR/VR), computer vision, statistical methods in AI